Programming Fundamentals

WEEK #2

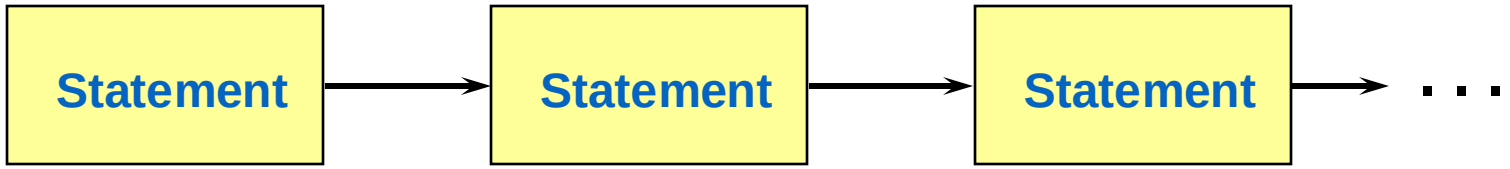
Arab Academy
for Science, Technology and Maritime Transport

Lecture Agenda

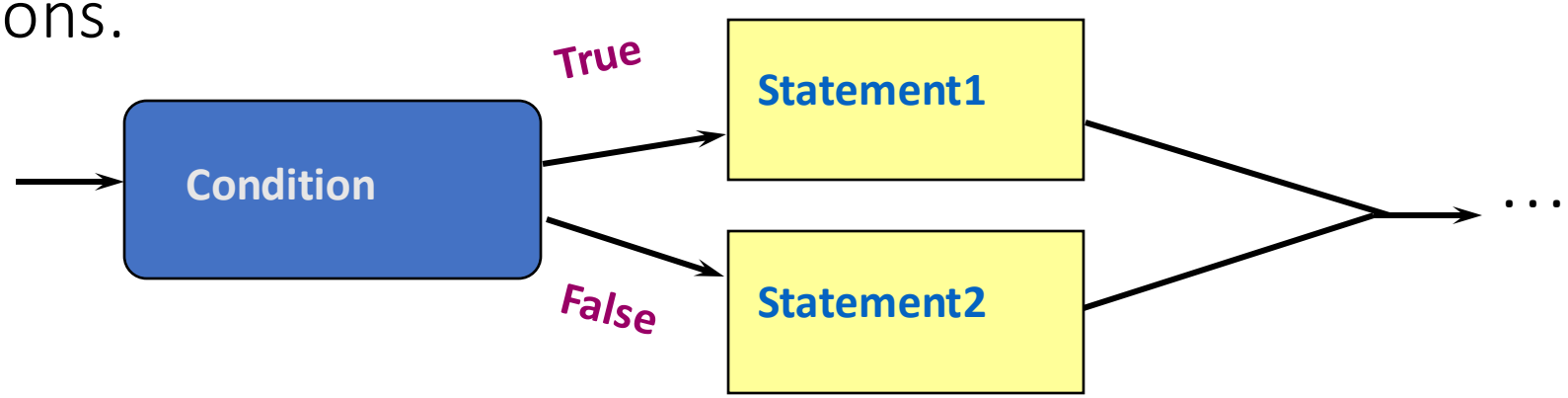
- **Program Designs**
- **Flow Charts**
- **Examples**

PROGRAM DESIGNS

a) Sequence: A sequence control structure is simply a series of procedures that follow one another.

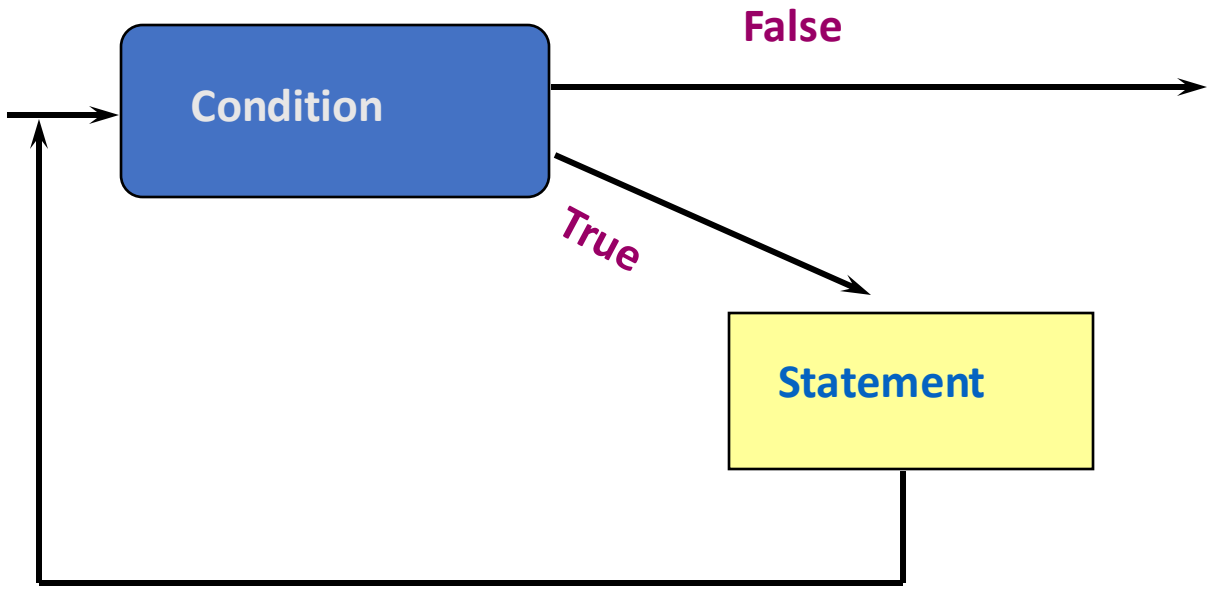


b) Selection: is used to execute different statements depending on certain conditions.



PROGRAM DESIGNS

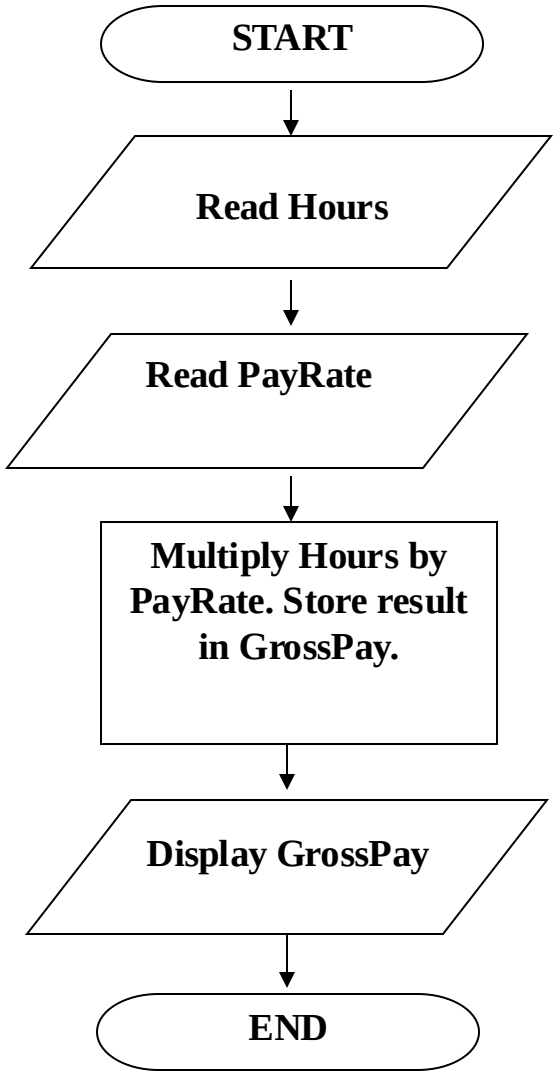
c) Iteration(repetition): is used to repeat statements while certain conditions are met.



Flow chart

Example: pay-calculating

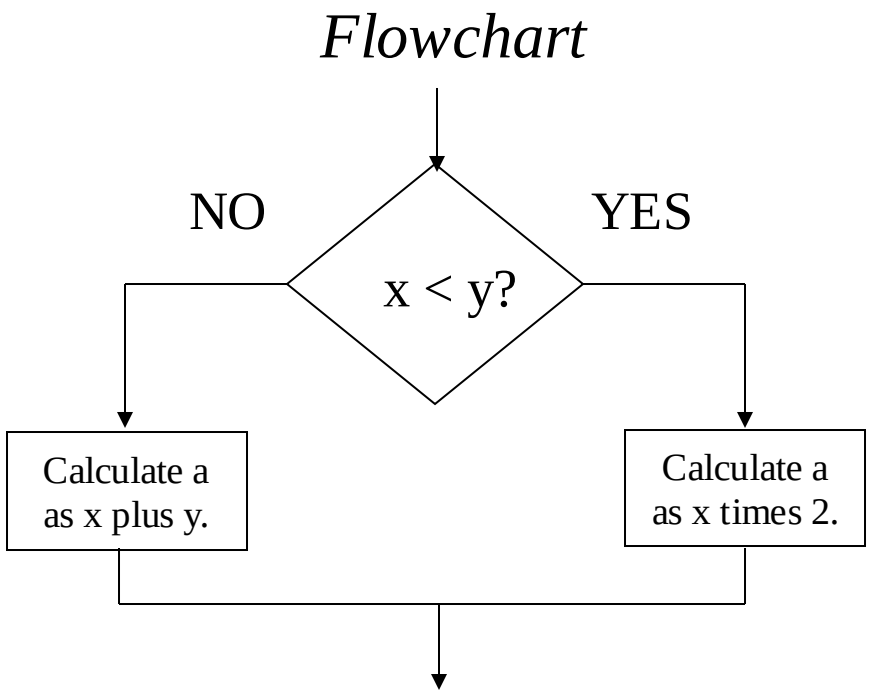
- The pay-calculating example was a sequence flowchart.



Flowchart

Decision Structure

- The flowchart segment below shows how a decision structure if/else statement.



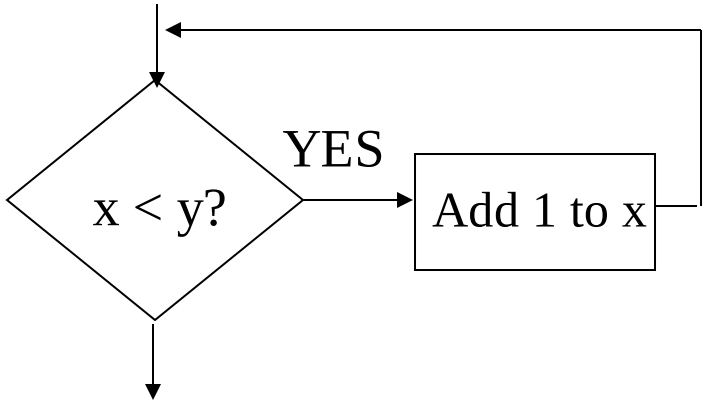

```
if (x < y)
    a = x * 2;
else
    a = x + y;
```

Flowchart

Repetition Structure

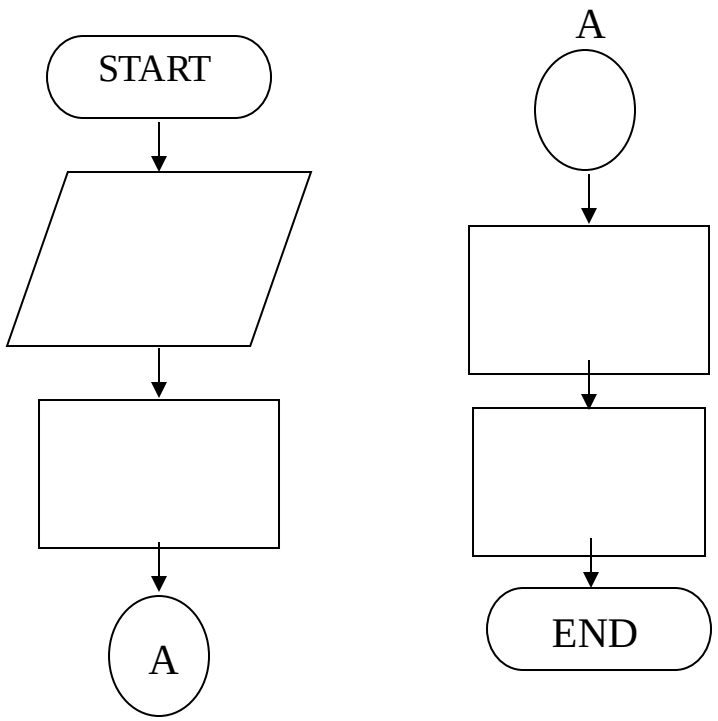
- The flowchart segment below shows a repetition structure expressed in Code a while loop.

Flowchart



Flowchart
Connectors

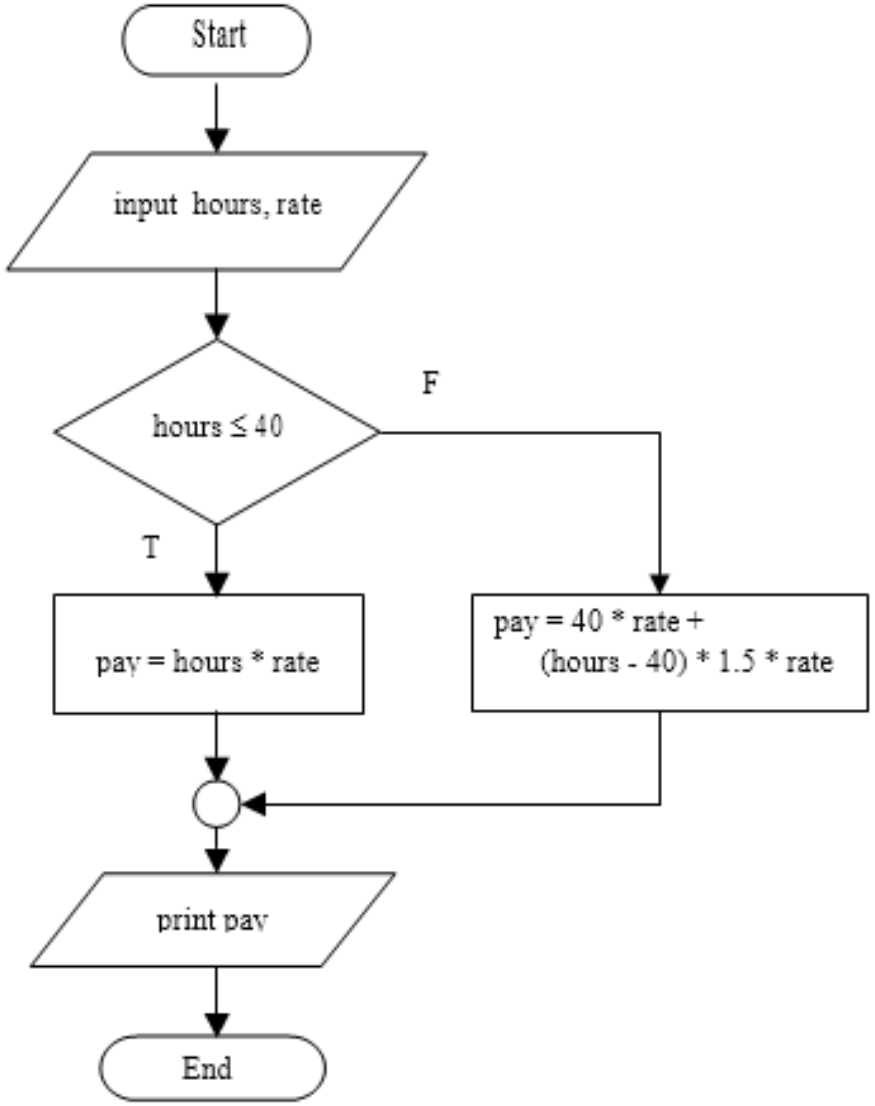
- The "A" connector indicates that the second flowchart segment begins where the first segment ends.



Examples

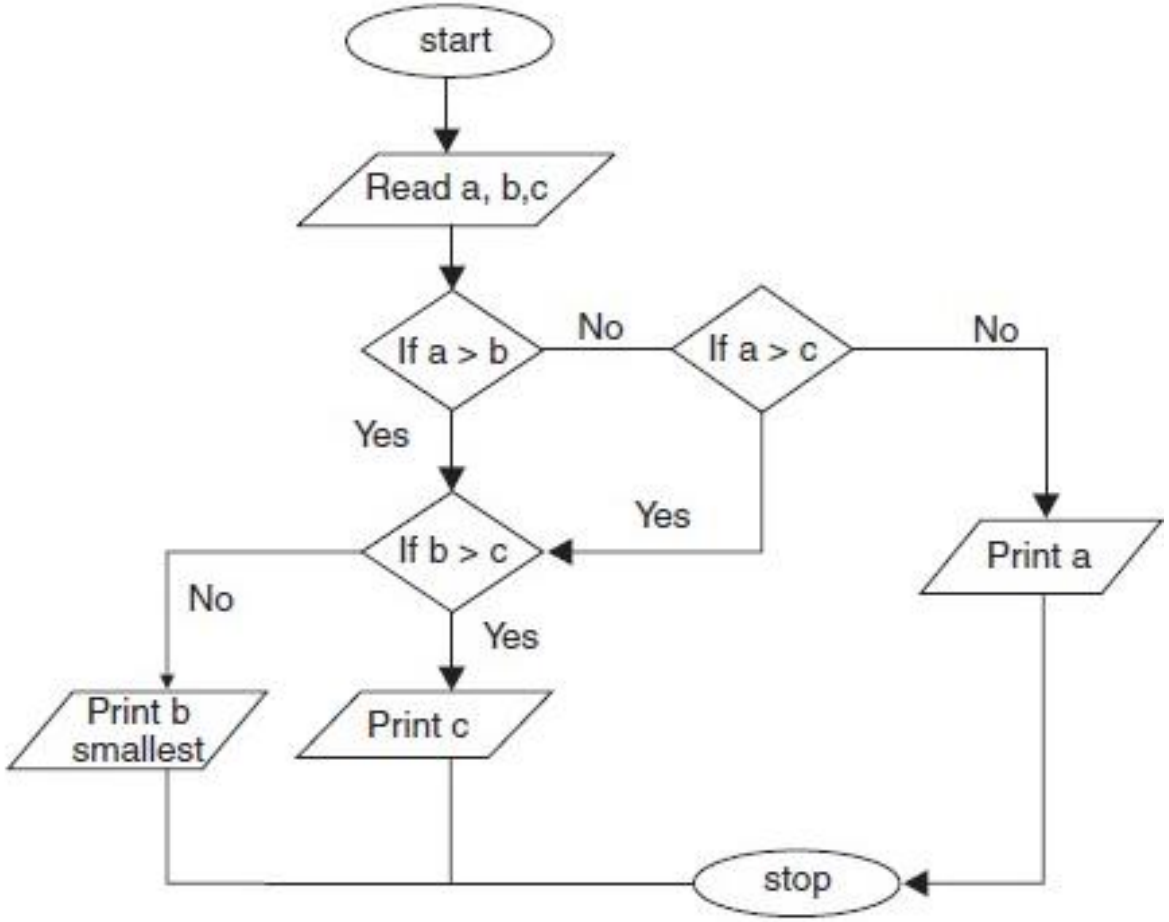
Example: Calculate Pay with Overtime - selection

```
Begin
    input hours, rate
    if hours <= 40 then
        pay =  hours * rate
    else
        pay =  40 * rate +  (hours – 40) * rate * 1.5
    print pay
End
```



Examples

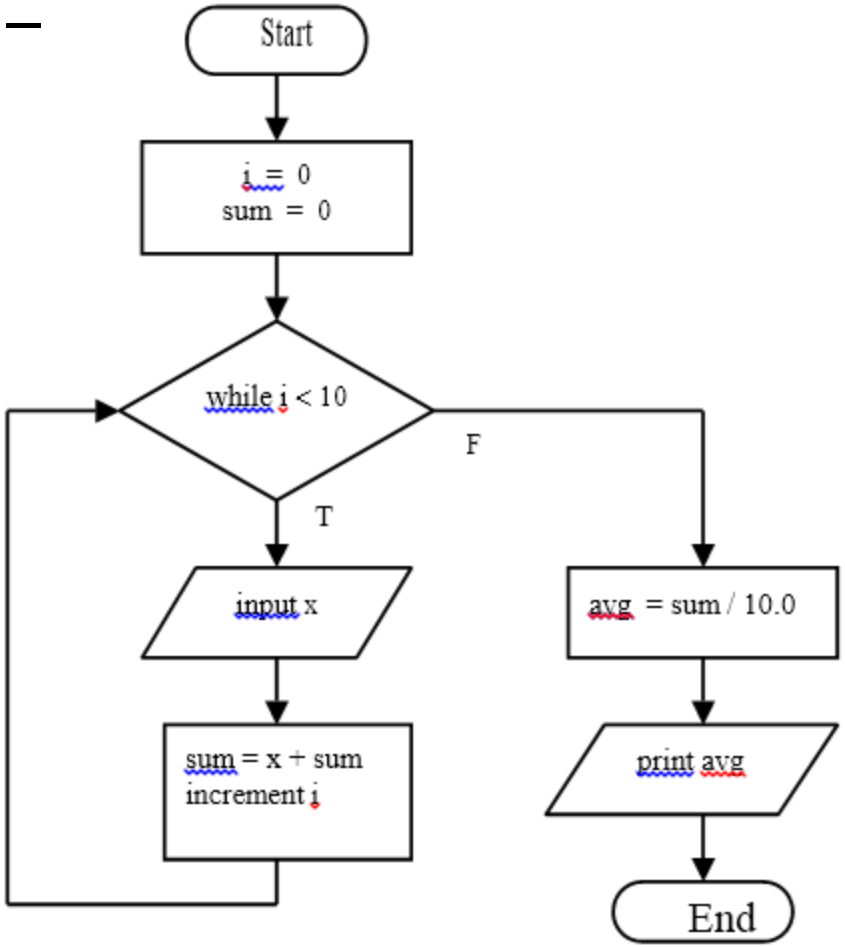
Example: flowchart to find the smallest among three different numbers



Examples

Example: Average of 10 Numbers – iteration with a while loop

```
Begin
    i  =  0
    sum  =  0
    while i < 10
        input x
        sum =  sum + x
        Increment i
    avg  =  sum / 10.0
    print avg
End
```

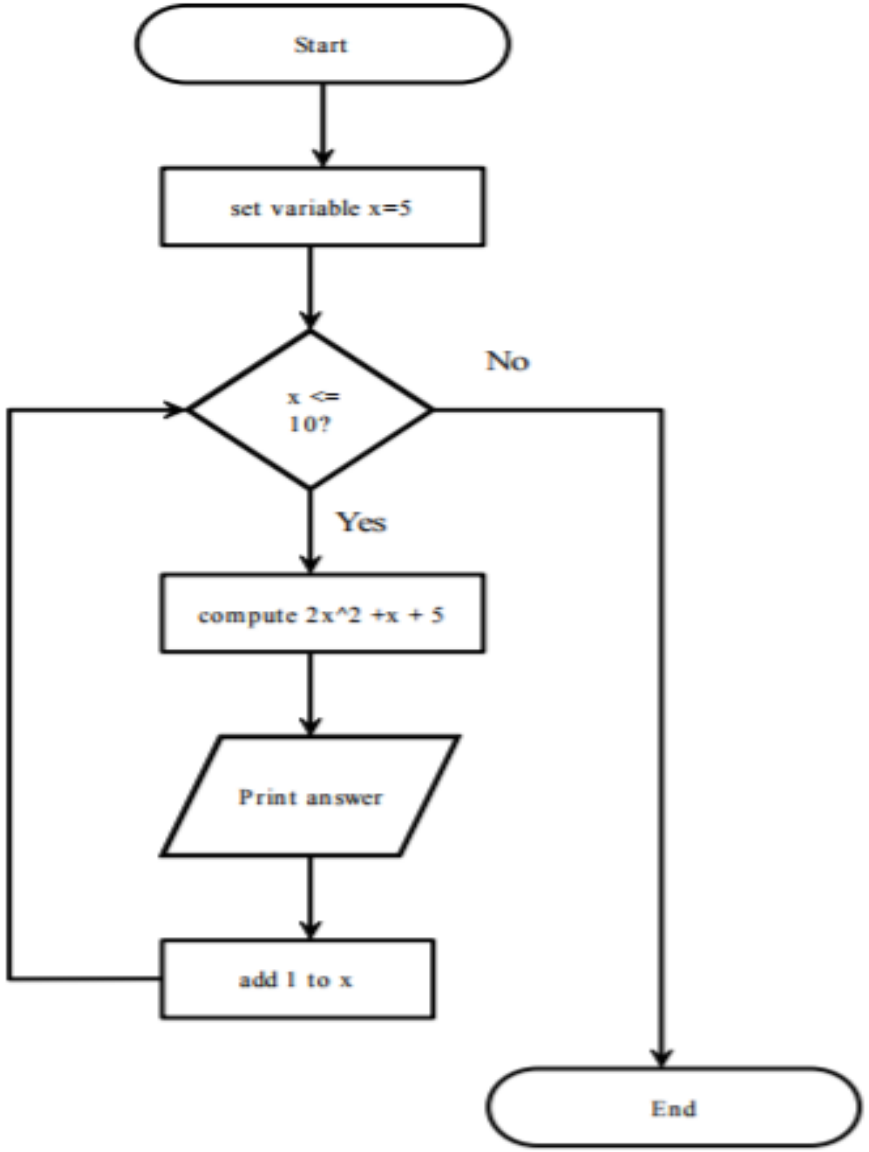


Examples

- Example: Solving the equation $2x^2 + x + 5$ for all x between 5 and 10

- The program should take x and start it at 5. Next, substitute 5 for each value of x in the equation and then solve (answer = 60). Now take the next x, which is obviously 6 (5 + 1). Solve the equation for x = 6 and continue on. The last value of x you will do this for is 10.

Examples




```
Set x to 5
While x <= 10
        Compute 2x^2 + x + 5
        Print answer
        Increment x
        Return to While Statement
Quit
```

Thank You

- Any Further Questions??